

Gretel Quintero Angulo

Ph.D. IN PHYSICS

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PROFILE

I am a curiosity-driven individual with a strong background in research and education in physics. After an extensive career in academia, I am seeking to expand my professional scope by discovering new and exciting environments where I can use my skills and acquire new ones.

EXPERIENCE

02.2023 – present **Alexander von Humboldt Postdoc Fellow**

Institut für Theoretische Physik I, Heinrich-Heine-Universität, Düsseldorf

- Mathematical modelling of axion-like excitations in topological insulators and dense quark matter under strong magnetic fields.
- Analytical and numerical general relativistic description of the hydrostatic equilibrium of magnetized stars.
- Presentation of the results at seminars and conferences.
- Scientific articles writing and peer-reviewing.
- Bachelor and Master thesis supervision.
- Scientific conferences organization.

02.2020 – 01.2023 **Associate Professor**

Department of Theoretical Physics, Faculty of Physics, University of Havana, Havana, Cuba

- Courses: Statistical Physics, Mathematical Methods for Physicists, Mathematical Analysis, Analytic Geometry.
- Courses and academic programs design.
- Mentoring of students and young professors.
- Head of the Department, from June 2021 to November 2022.
- Evaluation of doctoral master, and bachelor thesis.

- Evaluation of the candidates to the positions of Instructor and Assistant professors.
- Evaluation of the scientific projects of undergraduate students.
- Evaluation of the teaching work of the professors in training.

08.2019 – 02.2023 External associate Researcher

Department of Theoretical Physics, Institute of Cybernetics, Mathematics and Physics (ICIMAF), Havana, Cuba

- Analytical and numerical modeling of the physics of dense matter under strong magnetic fields in astrophysical environments.
- Presentation of the results at seminars and conferences.
- Scientific articles writing and peer-reviewing.
- Bachelor and Master thesis supervision.
- Scientific conferences organization.
- Scientific projects writing and evaluation.
- Science dissemination and outreach activities organization.

11.2016 – 07.2019 Ph.D. Student

Department of Theoretical Physics, Institute of Cybernetics, Mathematics and Physics (ICIMAF)

- Analytical and numerical modeling of the physics of dense matter under strong magnetic fields in astrophysical environments.
- Presentation of the results at seminars and conferences.
- Scientific articles writing and peer-reviewing.
- Scientific conferences organization.
- Science dissemination and outreach activities organization.

03.2013 – 01.2020 Assistant Professor

Department of Theoretical Physics, Faculty of Physics, University of Havana

- Courses: Statistical Physics, Mathematical Methods for Physicists and Engineers, Analytic Geometry.
- Courses and academic programs design.
- Evaluation of master and bachelor thesis.
- Evaluation of the scientific projects of undergraduate students.
- Evaluation of the assistant students and professors in training.

01.2010 – 02.2013 Instructor

Department of Theoretical Physics, Faculty of Physics, University of Havana

- Courses: Statistical Physics, Mathematical Methods for Physicists, Theoretical Mechanics.
- Master Student; preparation of the master dissertation.
- Courses and academic programs design.
- Evaluation of bachelor thesis.

09.2007 – 12.2009 Postgraduate teaching training

Department of Theoretical Physics, Faculty of Physics, University of Havana

- Courses: Mathematical Methods for Physicists, Mathematical Analysis, Laboratory of Mechanics and Optics.
- Member of the Admittance Committee of the Faculty of Physics (design and grading of the admittance test).

**Research
record:**

[ORCID: 0000-0002-7301-5276](#)
[Full list of publications](#)
[Full list of presentations](#)
[Awards and Honors](#)

Research areas: Astrophysics, Particle Physics, Statistical Physics, Condensed matter Physics

Memberships: Alexander von Humboldt Foundation
Cuban Physics Society

Referee at: Superconductivity, 2022
Modern Physics Letter A, 2019
Revista Cubana de Física (Cuban Physics Review), since 2017

**Training
activities:** Advised B.Sc. in Physics: Equations of Stellar Structure for Magnetized Compact Stars, Adriel Rodríguez Concepción Pérez, Faculty of Physics, University of Havana, finished in July 2024.

Co-advised M.Sc. in Physics: Thermodynamics of a neutral vector boson gas, Lismary de la Caridad Suárez González, University of Havana, finished in December 2020.

Co-advised B.Sc. in Physics: Computation of observables of magnetized strange stars, Samantha López Pérez, Faculty of Physics, University of Havana, finished in July 2020.

**Scientific
conferences
organization:**

Workshop on Heavy Ion Collisions and Compact Stars - Hipstars 2022. Hybrid Conferences/Havana time zone, October 24th-28th, 2022.

German-Cuban Summer School 2021. ICIMAF, Cuba-FIAS, Germany. January 17th-21st, 2022

FT-45, ICIMAF, Cuba. Online conference. Mars 14th -16th, 2021.

HIPSTARS-Workshop on Heavy Ion Physics and Compact Stars. Online conference. December, 1st -4th, 2020.

XV Symposium of the Cuban Physics Society. Havana, Cuba. Mars 9th-13rd, 2020.

Caribbean Symposium on Cosmology, Gravitation, Nuclear and Astroparticle Physics and International Symposium on Strong Electromagnetic Fields and Neutron Stars, (STARS/SMFNS 2019), Havana/Varadero, Cuba, May 6th-12th, 2019.

**Projects and
funding:**

Internationals:

Humboldt Research Fellowship for Postdocs. Institute for Theoretical Physics I, Heinrich Heine University, Düsseldorf, Germany. 01.02.2023 - 31.01.2025.

Grant of the Office of External Activities of the Abdus Salam International Centre for Theoretical Physics (ICTP), NT-09 (2018, 2019, 2020).

Grant of the Service for Cooperation and Cultural Action (SCAC) of the Embassy of France in Cuba for a one-month internship at LUTH, Paris Observatory, Paris, France. September 26th-October 26th, 2018.

Grant of the Federated Institutes Program of the ICTP for a one-month internship at the ICTP, Trieste, Italy. October 27th-November 28th, 2018.

Nationals:

Quantum and classical field theory at high energies, condensed matter, biophysics, and astroparticle physics. Project non-associated to Program PNAP-AENTA Cuba No.NA211LH500-022, 2021-2022 awarded to the DFT-ICIMAF (Participant.)

Quantum field theory and many-body theory in astrophysics, high-energy physics, and condensed matter physics. Project of the Cuban National Program for Basics Sciences PNCB-MES Cuba No. 500.03401, 2016-2020 awarded to the DFT-ICIMAF (Participant.)

EDUCATION

Academic Qualifications

11.2016 to 07.2019 **Ph.D. in Physics;** Institute of Cybernetics, Mathematics and Physics (ICIMAF-Cuba) and Faculty of Physics, University of Havana.
Thesis: Magnetic field effects in a gas of neutral vector bosons: Astrophysical applications

Summa Cum Laude. 5/5 points

01.2010 to 01.2013 **M.Sc. in Physics**; Faculty of Physics, University of Havana.
Thesis: Stability of the replica symmetric solution in diluted perceptron learning
Summa Cum Laude. 5/5 points

09.2002 to 07.2007 **B.Sc. in Physics**; Faculty of Physics, University of Havana.
Thesis: Skin effect in vapor-grown carbon fibers
Summa Cum Laude. 5/5 points

Supplementary education

04.2021 to 06.2021 **Diploma on Gender Theory and Communication**; International Institute of Journalism José Martí, Havana, Cuba.

09.2015 to 07.2016 **Diploma in Humanism and Society**; Alberto Hurtado University, Chile, and Fray Bartolomé de las Casas Study Center, Havana, Cuba.

10.2012 to 10.2013 **Course on Storytelling and Narrative Techniques**; Literary Training Center Onelio Jorge Cardoso, Havana Cuba.

SKILLS

Computer Skills: C, C++, Mathematica, Windows, Linux, Latex, Office

Soft Skills: Analytical thinking. Problem solving. Scientific research, education, communication and writing. Team management. Interdisciplinary teamwork. Mentoring and Coaching. Networking and Outreach. Empathy and Adaptability.

Languages: Spanish: Mother tongue English: Very good knowledge
French: Good knowledge German: Basic knowledge

OTHER ACTIVITIES AND INTERESTS

Outreach and volunteering: Organizer of the Workshops on gender and science as a celebration of the International Day of Women and Girls in Science: ICIMAF-Cuban Physics Society-Cuban Mathematics Society. February 11th, 2021, 2022, 2023.
Scientific advisor for the Cuban TV show "*Una calle, mil caminos*". Episode: "*La vida media del muón*". Broadcasted on September 2nd, 2021.
Founder and director of the theater group "*Monopolo Magnético*" (Magnetic Monopole), formed by students and professors of physics at the University of Havana, 2015-2020.

Co-founder and organizer of the Cuban Meeting for Graduate Students in Physics (FISESP), Havana, Cuba. First (19th-21th January, 2018); second (15th-18th January, 2020).

Organizer and lecturer at the First Summer School for high school students, Faculty of Physics, University of Havana, July 2019.

Member of the Cuban Steering Committee for the IAU100 NameExoWorlds Project as part of the International Astronomical Union's 100th-anniversary celebrations, 2019.

[Publications for science dissemination at the end here.](#)

Free Time Activities: Literature and writing. Movies, music and theater. Traveling and history. Spa and massage. Walking tours.

Personal Blog: elinsomniodegretel.de